

Chapter 3

Continuation of Equilibria

3.1 Parameterized Families of Equilibria

Radii polynomials are introduced in Chapter 2 as a means of rigorously verifying the existence of an equilibrium to an ODE of the form $\dot{x} = f(x)$. However, in applications parameters play a fundamental role and thus it is important to be able to find equilibria for

$$\dot{x} = f(x, \lambda),$$

where $f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $\lambda \in \mathbb{R}^m$ denotes the parameters. Assume that $f(x_0, \lambda_0) = 0$. An obvious question is under what conditions does there exist a neighborhood $V \subset \mathbb{R}^m$ of λ_0 and a continuous function $\alpha: V \rightarrow \mathbb{R}^n$ such that $\alpha(\lambda_0) = x_0$ and $f(\alpha(\lambda), \lambda) = 0$. The implicit function theorem provides an answer.

Theorem 3.1.1. [Implicit Function Theorem] *Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ be open sets. Assume that $f: U \times V \rightarrow \mathbb{R}^n$ is C^1 and that $f(x_0, y_0) = 0$ for $(x_0, y_0) \in U \times V$. If $D_x f(x_0, y_0)$ is invertible, then there exist neighborhoods $U_0 \subset U$ and $V_0 \subset V$ of x_0 and λ_0 , respectively, and a C^1 function $\alpha: V_0 \rightarrow U_0$ such that $f(\alpha(y), y) = 0$. Furthermore, if $(x, y) \in U_0 \times V_0$ and $f(x, y) = 0$, then $x = \alpha(y)$.*

We take a slightly different, more restrictive approach in this chapter and assume that $\lambda \in \mathbb{R}$. In this context, nondegenerate zeros of f are locally parameterized families over the one-dimensional parameter λ . In other words, solutions are branches (or curves). Let $\Lambda \subset \mathbb{R}$ and consider a C^1 map $f: \mathbb{R}^n \times \Lambda \rightarrow \mathbb{R}^n$ with $\Lambda \subset \mathbb{R}$. Denote the *solution set* by

$$\mathcal{S} \stackrel{\text{def}}{=} \{(x, \lambda) \in \mathbb{R}^n \times \mathbb{R} \mid f(x, \lambda) = 0\} \subset \mathbb{R}^n \times \mathbb{R}.$$

We briefly introduce two classical algorithms from numerical analysis to compute branches of solutions in $\mathcal{S}$. The first one is called parameter continuation and is introduced in Section 3.1.1, and the second one is called pseudo-arclength continuation and is presented in Section 3.1.2. These methods fall into the class of *predictor-corrector algorithms*.

3.1.1 Parameter Continuation

This method involves a predictor and a corrector step: given, within a prescribed tolerance, a solution x_0 at parameter value λ_0 , the predictor step produces an approximate solution $\hat{x}_0$ at nearby parameter value $\lambda_1 = \lambda_0 + \Delta_\lambda$ (for some $\Delta_\lambda \neq 0$), and the corrector step, takes $\hat{x}_1$ as its input and produces with Newton's method, once again within the prescribed tolerance, a solution x_1 at λ_1 .

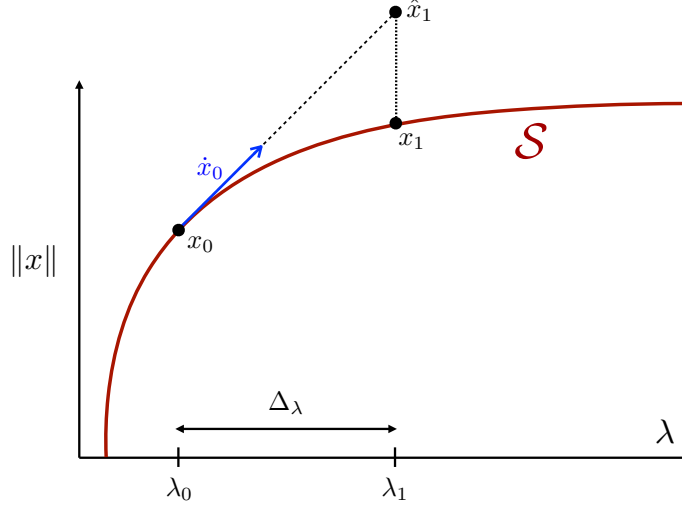


Figure 3.1: Parameter continuation.

The predictor is obtained by assuming that at the solution (x_0, λ_0) , the Jacobian matrix $D_x f(x_0, \lambda_0)$ is invertible, which in turns implies by the implicit function theorem (Theorem 3.1.1) that the solution curve is locally parametrized by λ . In this case, close to (x_0, λ_0) , we have

$$\frac{\partial}{\partial \lambda}(f(x, \lambda) = 0) \iff D_x f(x, \lambda) \frac{dx}{d\lambda}(\lambda) + \frac{\partial f}{\partial \lambda}(x, \lambda) = 0 \iff \frac{dx}{d\lambda}(\lambda) = -D_x f(x, \lambda)^{-1} \frac{\partial f}{\partial \lambda}(x, \lambda).$$

At (x_0, λ_0) , a tangent vector to the curve is $\dot{x}_0 \stackrel{\text{def}}{=} \frac{dx}{d\lambda}(\lambda_0)$ and is obtained with the formula

$$\dot{x}_0 = -D_x f(x_0, \lambda_0)^{-1} \frac{\partial f}{\partial \lambda}(x_0, \lambda_0).$$

Once the tangent vector $\dot{x}_0$ is obtained, the predictor is defined by

$$\hat{x}_1 = x_0 + \Delta_\lambda \dot{x}_0.$$

Then, fixing $\lambda_1 = \lambda_0 + \Delta_\lambda$, we *correct* the predictor $\hat{x}_1$ using Newton's method

$$x_1^{(0)} = \hat{x}_1, \quad x_1^{(n+1)} = x_1^{(n)} - \left(D_x f(x_1^{(n)}, \lambda_1) \right)^{-1} f(x_1^{(n)}, \lambda_1), \quad n \geq 0,$$

to obtain the solution x_1 at λ_1 within the prescribed tolerance. We repeat this procedure iteratively to produce numerically a branch of solutions. We refer to Figure 3.1 to visualize one step of the parameter continuation algorithm.

Parameter continuation, while simple and easy to use, has a major disadvantage: it will necessarily fail to converge close to degenerate zeros, as the Jacobian matrix at a degenerate zero is singular. This is for instance what is happening when a saddle-node bifurcations (folds) occur (see Section 4.6 for a proper mathematical definition). An example of such phenomenon is given by $f(x, \lambda) = x^2 - \lambda = 0$ at the point $(x_0, \lambda_0) = (0, 0)$. In order to address this issue, Keller (e.g. see Keller [21]) introduced pseudo-arclength continuation, which allows continuing past folds. Instead of parameterizing the branch of solutions by the natural parameter λ , it is parametrized by arclength or *pseudo-arclength*. The next section provides more details.

3.1.2 Pseudo-Arclength Continuation

In the pseudo-arclength continuation algorithm, the parameter value λ is no longer fixed and instead is left as a variable. The unknown variable is now $X = (x, \lambda)$. Consider the problem $f(X) = 0$ with the map $f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$. As before, the process begins with a solution $X_0 = (x_0, \lambda_0)$ given within a prescribed tolerance. To produce a predictor, we compute first a unit tangent vector to the curve at X_0 , that we denote $\dot{X}_0$, which is computed using the formula

$$D_X f(X_0) \dot{X}_0 = \left[D_x f(x_0, \lambda_0) \quad \frac{\partial f}{\partial \lambda}(x_0, \lambda_0) \right] \dot{X}_0 = 0 \in \mathbb{R}^n.$$

We now fix a *pseudo-arclength parameter* $\Delta_s > 0$, and set the predictor to be

$$\hat{X}_1 \stackrel{\text{def}}{=} \bar{X}_0 + \Delta_s \dot{X}_0 \in \mathbb{R}^{n+1}.$$

Once the predictor is fixed, we *correct* toward the set $\mathcal{S}$ on the hyperplane perpendicular to the tangent vector $\dot{X}_0$ that contains the predictor $\hat{X}_1$. The equation of this plan is given by $\{X \in \mathbb{R}^{n+1} \mid (X - \hat{X}_1) \cdot \dot{X}_0 = 0\}$. Define $E(X) \stackrel{\text{def}}{=} (X - \hat{X}_1) \cdot \dot{X}_0$ and then we can apply Newton's method to the new problem $F : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$ given by

$$F(X) \stackrel{\text{def}}{=} \begin{pmatrix} E(X) \\ f(X) \end{pmatrix} = 0, \quad (3.1)$$

with the initial condition $\hat{X}_1$ to obtain a new solution X_1 given again within a prescribed tolerance. See Figure 3.2 for a geometric interpretation of one step of the pseudo-arclength continuation algorithm. At each step of the algorithm, the function defined in (3.1) changes since the plane $E(X) = 0$ changes. With this method, it is possible to continue past folds. Repeating this procedure iteratively produces a branch of solutions.

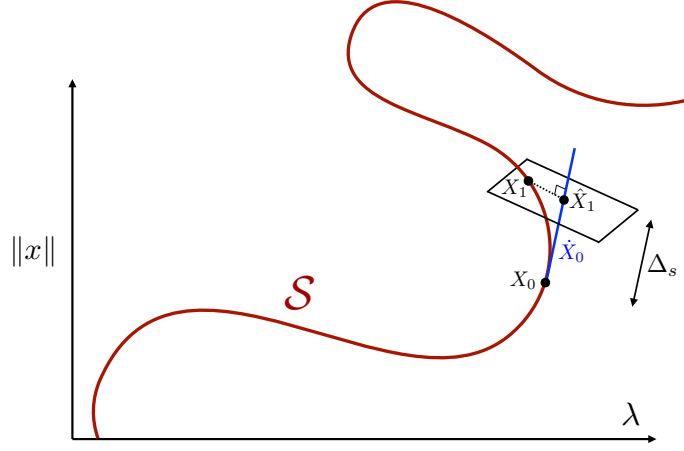


Figure 3.2: Pseudo-arclength continuation.

3.2 Rigorous Parameter Continuation for Zeros of Functions

Having introduced in Section 3.1 two classical predictor-corrector algorithms to compute numerically branches of solutions of $f(x, \lambda) = 0$, we turn to the question of proving our computations obtained from parameter continuation as introduced in Section 3.1.1. Assume we have approximate solutions to $f(x, \lambda) = 0$ given by $(\bar{x}_0, \lambda_0)$ and $(\bar{x}_1, \lambda_1)$. Let

$$\lambda_s \stackrel{\text{def}}{=} (1-s)\lambda_0 + s\lambda_1 \quad \text{and} \quad \bar{x}_s \stackrel{\text{def}}{=} (1-s)\bar{x}_0 + s\bar{x}_1, \quad s \in [0, 1]. \quad (3.2)$$

We call the set of points $\{(\bar{x}_s, \lambda_s) \mid s \in [0, 1]\}$ the *predictors*. See Figure 3.3 for a geometric representation of the predictors.

We want to use radii polynomials to conclude about the existence of a smooth curve of equilibria $\tilde{x}_s$ lying in a region centered about the segment $\{\bar{x}_s \mid s \in [0, 1]\}$. In Chapter 2, the contraction mapping theorem is used, via the radii polynomial approach, to prove existence of zeros of functions at fixed parameter values. The following concept will allow adapting the radii polynomial approach to prove existence of branches of zeros.

Definition 3.2.1. A function $\mathcal{T}: X \times \Lambda \rightarrow X$ is a *uniform contraction* over Λ if there exists $\kappa \in [0, 1)$ such that, for all $x, y \in X$ and $\lambda \in \Lambda$,

$$\|\mathcal{T}(x, \lambda) - \mathcal{T}(y, \lambda)\|_X \leq \kappa \|x - y\|_X.$$

By the contraction mapping theorem (see Theorem 2.1.3) if $\mathcal{T}: X \times \Lambda \rightarrow X$ is a uniform contraction, then for every $\lambda \in \Lambda$ there exists a unique $\tilde{x}_\lambda$ such that $\mathcal{T}(\tilde{x}_\lambda, \lambda) = \tilde{x}_\lambda$. Thus the function $g: \Lambda \rightarrow X$ given by $g(\lambda) \stackrel{\text{def}}{=} \tilde{x}_\lambda$ is well defined. As the following theorem indicates this function inherits the same amount of differentiability than T . The proof is found in [11].

Theorem 3.2.2 (Uniform Contraction Theorem). *Assume that X and Λ are Banach spaces. Consider open sets $U \subset X$ and $V \subset \Lambda$. Assume that $\mathcal{T}: \bar{U} \times V \rightarrow \bar{U}$ is a uniform contraction with contraction constant κ . Define $g: V \rightarrow \bar{U}$ by $\mathcal{T}(g(\lambda), \lambda) = g(\lambda)$. If $\mathcal{T} \in C^k(\bar{U} \times V, X)$, then $g \in C^k(V, X)$ for any $k \in \{1, 2, \dots, \infty\}$. If $\mathcal{T}$ is analytic, then g is also analytic.*

The following theorem is a direct generalization of Theorem 2.4.2, and its proof is based on the uniform contraction theorem.

Theorem 3.2.3. *Consider $f \in C^k(\mathbb{R}^n \times \mathbb{R}, \mathbb{R}^n)$ with $k \in \{2, \dots, \infty, \omega\}$, and consider non negative bounds $Y_0, \hat{Y}_0, Z_0, \hat{Z}_0, Z_2$ satisfying*

$$\|Af(\bar{x}_0, \lambda_0)\| \leq Y_0, \quad (3.3)$$

$$\|A[f(\bar{x}_s, \lambda_s) - f(\bar{x}_0, \lambda_0)]\| \leq \hat{Y}_0, \quad \text{for all } s \in [0, 1] \quad (3.4)$$

$$\|Id - AD_x f(\bar{x}_0, \lambda_0)\| \leq Z_0, \quad (3.5)$$

$$\|A[D_x f(\bar{x}_s, \lambda_s) - D_x f(\bar{x}_0, \lambda_0)]\| \leq \hat{Z}_0, \quad \text{for all } s \in [0, 1] \quad (3.6)$$

$$\|A[D_x f(c, \lambda_s) - D_x f(\bar{x}_s, \lambda_s)]\| \leq Z_2(r)r, \quad \text{for all } c \in \overline{B_r(\bar{x}_s)} \text{ and } s \in [0, 1]. \quad (3.7)$$

Define the radii polynomial

$$p(r) \stackrel{\text{def}}{=} Z_2(r)r^2 - (1 - Z_0 - \hat{Z}_0)r + Y_0 + \hat{Y}_0. \quad (3.8)$$

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then there exists a function

$$\tilde{x}: [0, 1] \rightarrow \bigcup_{s \in [0, 1]} \overline{B_{r_0}(\bar{x}_s)}$$

with $\tilde{x} \in C^k((0, 1), \mathbb{R}^n)$, and such that

$$f(\tilde{x}(s), \lambda_s) = 0, \quad \text{for all } s \in [0, 1].$$

Furthermore, these are the only solutions in the “tube” $\bigcup_{s \in [0, 1]} \overline{B_{r_0}(\bar{x}_s)} \times \{\lambda_s\}$.

Proof. The assumption that $p(r_0) = (Z_2(r_0)r_0 + Z_0 + \hat{Z}_0)r_0 - r_0 + Y_0 + \hat{Y}_0 < 0$ leads to

$$\kappa \stackrel{\text{def}}{=} Z_0 + \hat{Z}_0 + Z_2(r_0)r_0 < 1 - \frac{Y_0 + \hat{Y}_0}{r_0} \leq 1. \quad (3.9)$$

Define the Newton-like operator $T: \mathbb{R}^n \times [0, 1] \rightarrow \mathbb{R}^n$ by

$$T(x, s) = x - Af(x, \lambda_s).$$

We begin by showing that for each $s \in [0, 1]$, the operator $T(\cdot, s)$ is a contraction mapping from $\overline{B_{r_0}(\bar{x}_s)}$ into itself. Now, given $y \in \overline{B_{r_0}(\bar{x}_s)}$ and applying the bounds (3.5), (3.6) and (3.7), we obtain

$$\begin{aligned} \|D_x T(y, s)\| &= \|Id - AD_x f(y, \lambda_s)\| \\ &\leq \|Id - AD_x f(\bar{x}_0, \lambda_0)\| + \|A[D_x f(\bar{x}_0, \lambda_0) - D_x f(\bar{x}_s, \lambda_s)]\| \\ &\quad + \|A[D_x f(y, \lambda_s) - D_x f(\bar{x}_s, \lambda_s)]\| \\ &\leq Z_0 + \hat{Z}_0 + Z_2(r_0)r_0 = \kappa < 1, \end{aligned} \tag{3.10}$$

where we applied (3.9) to obtain the strict inequality. We now show that for each $s \in [0, 1]$ the operator $T(\cdot, s)$ maps $\overline{B_{r_0}(\bar{x}_s)}$ into itself. Let $y \in \overline{B_{r_0}(\bar{x}_s)}$ and apply the Mean Value Inequality (Corollary 2.2.6) to obtain

$$\begin{aligned} \|T(y, s) - \bar{x}_s\| &\leq \|T(y, s) - T(\bar{x}_s, s)\| + \|T(\bar{x}_s, s) - \bar{x}_s\| \\ &\leq \sup_{b \in \overline{B_{r_0}(\bar{x}_s)}} \|D_x T(b, s)\| \|y - \bar{x}_s\| + \|Af(\bar{x}_s, \lambda_s)\| \\ &\leq \sup_{b \in \overline{B_{r_0}(\bar{x}_s)}} \|D_x T(b, s)\| \|y - \bar{x}_s\| + \|Af(\bar{x}_0, \lambda_0)\| + \|A[f(\bar{x}_s, \lambda_s) - f(\bar{x}_0, \lambda_0)]\| \\ &\leq (Z_0 + \hat{Z}_0 + Z_2(r_0)r_0)r_0 + Y_0 + \hat{Y}_0 \\ &< r_0, \end{aligned}$$

where the penultimate inequality follows from (3.10) and the last inequality follows from $p(r_0) < 0$. Hence, for all $y \in \overline{B_{r_0}(\bar{x}_s)}$ and $s \in [0, 1]$, $T(y, s) \in B_{r_0}(\bar{x}_s) \subset \overline{B_{r_0}(\bar{x}_s)}$.

Letting $a, b \in \overline{B_{r_0}(\bar{x}_s)}$, apply (3.10) to obtain

$$\begin{aligned} \|T(a, s) - T(b, s)\| &\leq \sup_{b \in \overline{B_{r_0}(\bar{x}_s)}} \|D_x T(b, s)\| \|a - b\| \\ &\leq (Z_0 + \hat{Z}_0 + Z_2(r_0)r_0) \|a - b\| \\ &= \kappa \|a - b\|. \end{aligned} \tag{3.11}$$

Define the operator

$$\begin{aligned} \mathcal{T} : \overline{B_{r_0}(0)} \times [0, 1] &\rightarrow \overline{B_{r_0}(0)} \\ (y, s) &\mapsto \mathcal{T}(y, s) \stackrel{\text{def}}{=} T(y + \bar{x}_s, s) - \bar{x}_s. \end{aligned}$$

Consider now $x, y \in \overline{B_{r_0}(0)}$ and $s \in [0, 1]$. Then, since $x + \bar{x}_s, y + \bar{x}_s \in \overline{B_{r_0}(\bar{x}_s)}$, we use (3.11) to get

$$\|\mathcal{T}(x, s) - \mathcal{T}(y, s)\| = \|T(x + \bar{x}_s, s) - T(y + \bar{x}_s, s)\| \leq \kappa \|x - y\|, \quad \text{for all } s \in [0, 1].$$

Since $\kappa < 1$, we conclude that $\mathcal{T} : \overline{B_{r_0}(0)} \times [0, 1] \rightarrow \overline{B_{r_0}(0)}$ is a uniform contraction. By the uniform contraction theorem (Theorem 3.2.2), there exists $g : [0, 1] \rightarrow \overline{B_{r_0}(0)}$ such that

$$\mathcal{T}(g(s), s) = g(s).$$

Since $f \in C^k(\mathbb{R}^n \times (0, 1), \mathbb{R}^n)$, then $\mathcal{T} \in C^k(\overline{B_{r_0}(0)} \times (0, 1), \overline{B_{r_0}(0)})$, and therefore $g \in C^k((0, 1), \overline{B_{r_0}(0)})$. Let

$$\tilde{x}(s) \stackrel{\text{def}}{=} g(s) + \bar{x}_s$$

so that for all $s \in [0, 1]$

$$T(\tilde{x}(s), s) = T(g(s) + \bar{x}_s, s) = \mathcal{T}(g(s), s) + \bar{x}_s = g(s) + \bar{x}_s = \tilde{x}(s).$$

Note that $\tilde{x} \in C^k((0, 1), \bigcup_{s \in [0, 1]} \overline{B_{r_0}(\bar{x}_s)})$. Since $T(x, s) = x - Af(x, \lambda_s)$, we get that

$$T(\tilde{x}(s), s) = \tilde{x}(s) - Af(\tilde{x}(s), \lambda_s) = \tilde{x}(s).$$

By (3.10), $Z_0 = \|Id - AD_x f(\bar{x}_0, \lambda_0)\| < 1$, which implies that A is an injective matrix. Hence,

$$f(\tilde{x}(s), \lambda_s) = 0, \quad \forall s \in [0, 1].$$

Furthermore, it follows from the contraction mapping theorem that $\tilde{x}(s)$ ($s \in [0, 1]$) are the only solutions in the tube $\bigcup_{s \in [0, 1]} \overline{B_{r_0}(\bar{x}_s)} \times [\lambda_0, \lambda_1]$. $\square$

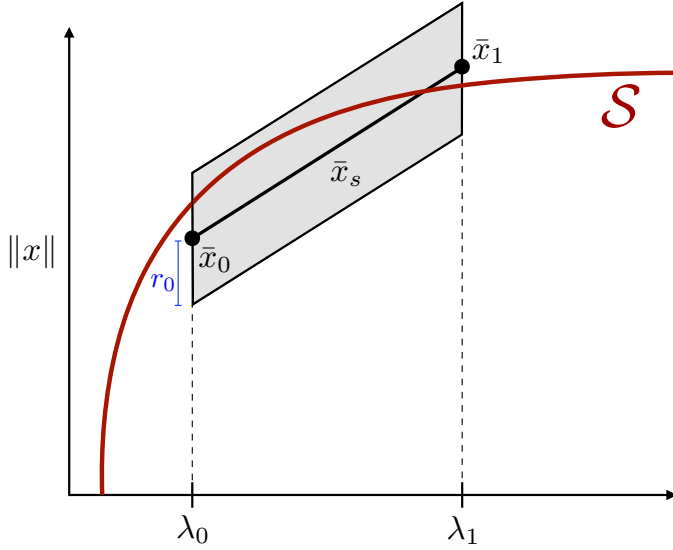


Figure 3.3: The set of predictors $\{(\bar{x}_s, \lambda_s) \mid s \in [0, 1]\}$, approximating a segment of the solution set $\mathcal{S}$. The radii polynomial approach, when successful, provides a “tube” of width $r_0 > 0$ (the shaded region) in $\mathbb{R}^n \times \mathbb{R}$, where a true segment of solution curve is guaranteed to exist.

3.2.1 Global Smooth Branches of Solutions Parameterized by λ

Theorem 3.2.3 provides a recipe to compute a local segment of solution curve and to obtain a uniform rigorous error bound r along the set of predictors connecting two numerical approximations $\bar{x}_0$ and $\bar{x}_1$. See Figure 3.3 for a representation of the region (shaded) where the true segment of solution curve is guaranteed to exist. Assume now that this argument has been repeated iteratively over the set $\{\bar{x}_0, \dots, \bar{x}_j\}$ of approximations at the parameter values $\{\lambda_0, \dots, \lambda_j\}$ respectively. For each $i = 0, \dots, j-1$, this yields the existence of a unique portion of smooth solution curve $\mathcal{S}_i$ in a small *tube* centered at the segment $\{(1-s)\bar{x}_i + s\bar{x}_{i+1} \mid s \in [0, 1]\}$. As the following results demonstrates, the set

$$\mathcal{S} \stackrel{\text{def}}{=} \bigcup_{i=0}^{j-1} \mathcal{S}_i$$

is a global smooth solution curve of $f(x, \lambda) = 0$.

Lemma 3.2.4 (Globalizing the C^k solution branch). *Assume that the radii polynomial approach was successfully applied (via Theorem 3.2.3) to show the existence of two C^k segments $\mathcal{S}_0$ and $\mathcal{S}_1$ of solution curves parameterized by the parameter λ over the respective parameter intervals $[\lambda_0, \lambda_1]$ and $[\lambda_1, \lambda_2]$. Assume that the sets of predictors are defined by the three points $\bar{x}_0$, $\bar{x}_1$ and $\bar{x}_2$. Then the new segment of solution curve $\mathcal{S}_0 \cup \mathcal{S}_1$ is a C^k function of λ .*

Proof. The continuity of $\mathcal{S}_0 \cup \mathcal{S}_1$ follows from the fact that at the parameter value $\lambda = \lambda_1$, the solution segment $\mathcal{S}_0$ must connect continuously with the solution segment $\mathcal{S}_1$ by the existence and uniqueness result guaranteed by the Contraction Mapping Theorem. Let $p_0(r)$ the radii polynomial built with the predictors generated by $\bar{x}_0$ and $\bar{x}_1$, and defined by the bounds Y_0 , $\hat{Y}_0$, Z_0 , $\hat{Z}_0$ and Z_2 . Let $r_0 > 0$ such that $p_0(r_0) < 0$. By continuity of the radii polynomial p_0 , there exists $\delta_0 > 0$ and there exist bounds $\tilde{Y}_0(\delta_0)$, $\tilde{Z}_0(\delta_0)$ and $\tilde{Z}_2(r, \delta_0)$ such that

$$\begin{aligned} \|A[f(\bar{x}_s, \lambda_s) - f(\bar{x}_0, \lambda_0)]\| &\leq \tilde{Y}_0(\delta_0), \quad \text{for all } s \in [-\delta_0, 1 + \delta_0] \\ \|A[D_x f(\bar{x}_s, \lambda_s) - D_x f(\bar{x}_0, \lambda_0)]\| &\leq \tilde{Z}_0(\delta_0), \quad \text{for all } s \in [-\delta_0, 1 + \delta_0] \\ \|A[D_x f(c, \lambda_s) - D_x f(\bar{x}_s, \lambda_s)]\| &\leq \tilde{Z}_2(r, \delta_0)r, \quad \text{for all } c \in \overline{B_r(\bar{x}_s)} \text{ and } s \in [-\delta_0, 1 + \delta_0], \end{aligned}$$

and such that

$$\tilde{p}_0(r_0) \stackrel{\text{def}}{=} \tilde{Z}_2(r_0, \delta_0)r_0^2 - \left(1 - Z_0 - \tilde{Z}_0(\delta_0)\right)r_0 + Y_0 + \tilde{Y}_0(\delta_0) < 0.$$

Then, there exists of a C^k branch of solution curve parameterized by λ over the range $\{(1-s)\lambda_0 + s\lambda_1 \mid s \in [-\delta_0, 1 + \delta_0]\}$, extending smoothly (in fact in a C^k way) the segment $\mathcal{S}_0$ on both sides. Similarly, there exists δ_1 such that the segment $\mathcal{S}_1$ is extended smoothly over the parameter range $\{(1-s)\lambda_1 + s\lambda_2 \mid s \in [-\delta_0, 1 + \delta_0]\}$. This implies that there is a C^k overlap between $\mathcal{S}_0$ and $\mathcal{S}_1$. $\square$

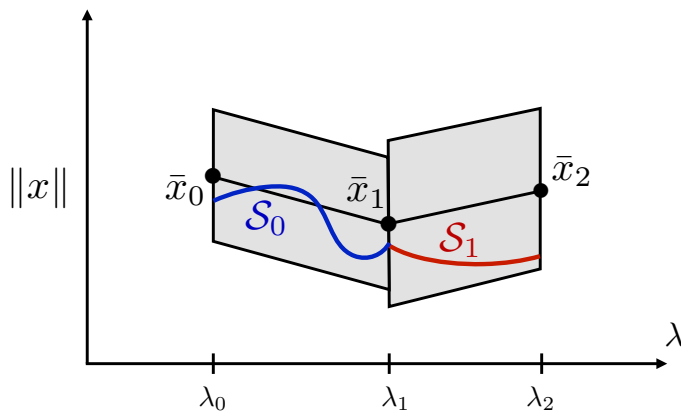


Figure 3.4: Assume that $\mathcal{S}_0$ and $\mathcal{S}_1$ are computed with the radii polynomial approach with predictors defined by three points $\bar{x}_0$, $\bar{x}_1$ and $\bar{x}_2$. Then the following situation is not possible: a piecewise smooth but not globally smooth piece of solution curve.

Repeating iteratively the argument of Lemma 3.2.4 leads to the existence of a smooth solution curve $\mathcal{S}$ of $f(x, \lambda) = 0$ near the piecewise linear curve of approximations, as portrayed in Figure 3.7.

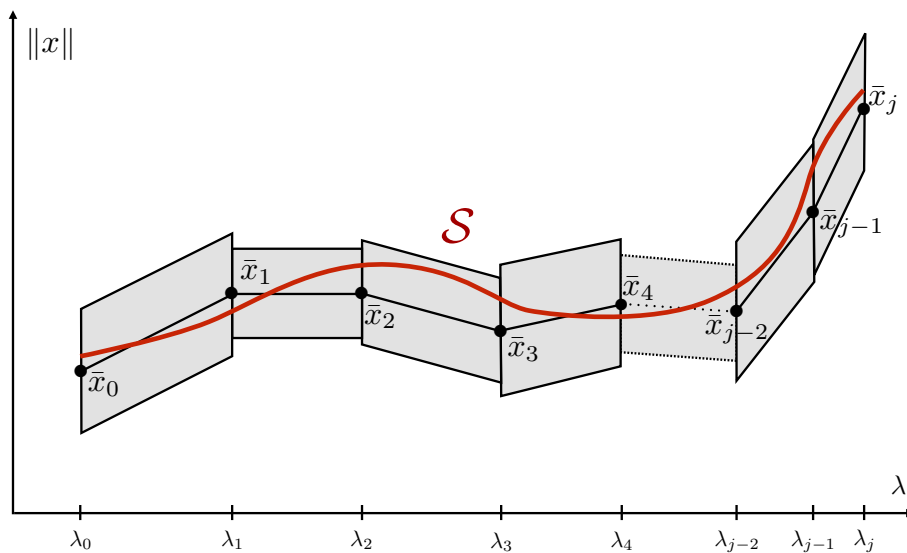


Figure 3.5: Computing a global branch of solutions with rigorous parameter continuation.

Example 3.2.5. As presented in more details in Section 13.1, the following example arises

when studying the dynamics of Fisher's partial differential equation model. Consider a number $N \in \mathbb{N}$ (the finite dimensional projection), and define $f : \mathbb{R}^{N+1} \times \mathbb{R} \rightarrow \mathbb{R}^{N+1}$ component-wise by

$$\dot{x}_k = f_k(x, \lambda) \stackrel{\text{def}}{=} (\lambda - k^2)x_k - \lambda(x^2)_k, \quad k = 0, \dots, N,$$

where λ is a parameter, and where

$$(x^2)_k = \sum_{\substack{k_1+k_2=k \\ k_i=-N, \dots, N}} x_{|k_1|}x_{|k_2|}, \quad k = 0, \dots, N.$$

For instance, $N = 1$ leads to the two-dimensional ODE

$$\dot{x} = f(x, \lambda) = \begin{pmatrix} \lambda x_0 - \lambda(x_0^2 + 2x_1^2) \\ (\lambda - 1)x_1 - \lambda(2x_0x_1) \end{pmatrix},$$

for $N = 2$ leads to the three-dimensional ODE

$$\dot{x} = f(x, \lambda) = \begin{pmatrix} \lambda x_0 - \lambda(x_0^2 + 2x_1^2 + 2x_2^2) \\ (\lambda - 1)x_1 - \lambda(2x_0x_1 + 2x_1x_2) \\ (\lambda - 4)x_2 - \lambda(2x_0x_2 + x_1^2) \end{pmatrix},$$

for $N = 3$ leads to the four-dimensional ODE

$$\dot{x} = f(x, \lambda) = \begin{pmatrix} \lambda x_0 - \lambda(x_0^2 + 2x_1^2 + 2x_2^2 + 2x_3^2) \\ (\lambda - 1)x_1 - \lambda(2x_0x_1 + 2x_1x_2 + 2x_1x_3) \\ (\lambda - 4)x_2 - \lambda(2x_0x_2 + x_1^2 + 2x_1x_3) \\ (\lambda - 9)x_3 - \lambda(2x_0x_3 + 2x_1x_2) \end{pmatrix}.$$

We now compute branches of equilibria for $\dot{x} = f(x, \lambda)$ by applying the technique of rigorous parameter continuation. For sake of generality, we construct the radii polynomial for a general finite dimensional projection N . For the norm, we choose the weighted 1-norm in $\mathbb{R}^{N+1}$ given by

$$\|x\|_1 = |x_0| + 2 \sum_{k=1}^N |x_k|.$$

The reason of choosing this norm is that the space $\mathbb{R}^{N+1}$ with the discrete convolution $x * y$ (with $x, y \in \mathbb{R}^N$) defined component-wise by

$$(x * y)_k = \sum_{\substack{k_1+k_2=k \\ k_i=-N, \dots, N}} x_{|k_1|}y_{|k_2|}, \quad k = 0, \dots, N$$

satisfies the property (2.55), that is $\|x * y\|_1 \leq \|x\|_1 \|y\|_1$, for all $x, y \in \mathbb{R}^{N+1}$.

Assume that at the parameter values $\lambda = \lambda_0$ and $\lambda = \lambda_1$ we computed $\bar{x}_0, \bar{x}_1 \in \mathbb{R}^{N+1}$ such that $f(\bar{x}_0, \lambda_0) \approx 0$ and $f(\bar{x}_1, \lambda_1) \approx 0$. Set $\Delta\bar{x} \stackrel{\text{def}}{=} \bar{x}_1 - \bar{x}_0$ and $\Delta\lambda \stackrel{\text{def}}{=} \lambda_1 - \lambda_0$. Compute a matrix $A \in M_{N+1}(\mathbb{R})$ such that $A \approx D_x f(\bar{x}_0, \lambda_0)^{-1}$. Compute Y_0 such that $\|A f(\bar{x}_0, \lambda_0)\|_1 \leq Y_0$. Recalling the $\hat{Y}_0$ bound satisfying (3.4). Given $s \in [0, 1]$, denote $X_s = (\bar{x}_s, \lambda_s) \in \mathbb{R}^{N+2}$ and $\Delta X = (\Delta\bar{x}, \Delta\lambda) = X_1 - X_0 = (\bar{x}_1 - \bar{x}_0, \lambda_1 - \lambda_0)$.

Denote by $K \in M_{N+1}(\mathbb{R})$ the diagonal matrix defined by $(Kx)_k = k^2 x_k$ so that we can conveniently write

$$f(x, \lambda) = (\lambda I - K)x - \lambda x^2. \quad (3.12)$$

Then, since there is a *cubic* term λx^2 (its Taylor expansion stops at order 3),

$$\begin{aligned} f(\bar{x}_s, \lambda_s) - f(\bar{x}_0, \lambda_0) &= f(X_s) - f(X_0) \\ &= (D_X f(X_0) \Delta X) s + \left(\frac{1}{2} D_X^2 f(X_0) (\Delta X, \Delta X) \right) s^2 \\ &\quad + \left(\frac{1}{6} D_X^3 f(X_0) (\Delta X, \Delta X, \Delta X) \right) s^3 \\ &=: y_1 s + y_2 s^2 + y_3 s^3, \end{aligned}$$

where $D_X f = \left[D_x f \frac{\partial f}{\partial \lambda} \right]$, is an $(N+1) \times (N+2)$ dimensional matrix, and where

$$\begin{aligned} y_1 &= (\lambda_0 I - K) \Delta\bar{x} - 2\lambda_0 (\bar{x}_0 * \Delta\bar{x}) + (\bar{x}_0 - \bar{x}_0^2) \Delta\lambda \\ y_2 &= -(\Delta\bar{x} * \Delta\bar{x}) \lambda_0 - 2(\bar{x}_0 * \Delta\bar{x}) \Delta\lambda + \Delta\bar{x} \Delta\lambda \\ y_3 &= -(\Delta\bar{x} * \Delta\bar{x}) \Delta\lambda. \end{aligned}$$

This expansion allows computing $\hat{Y}_0$ such that

$$\|A[f(\bar{x}_s, \lambda_s) - f(\bar{x}_0, \lambda_0)]\|_1 \leq \|Ay_1\|_1 + \|Ay_2\|_1 + \|Ay_3\|_1 \leq \hat{Y}_0.$$

Compute Z_0 such that $\|I - A D_x f(\bar{x}_0, \lambda_0)\|_1 \leq Z_0$. Recalling (3.6), we now look for a bound $\hat{Z}_0$ satisfying $\|A[D_x f(\bar{x}_s, \lambda_s) - D_x f(\bar{x}_0, \lambda_0)]\|_1 \leq \hat{Z}_0$, for all $s \in [0, 1]$. Given $h \in \overline{B_1(0)}$

$$D_x f(\bar{x}_s, \lambda_s) h - D_x f(\bar{x}_0, \lambda_0) h = (\Delta\lambda h - 2\Delta\lambda(\bar{x}_0 * h) - 2\lambda_0(\Delta\bar{x} * h)) s - 2\Delta\lambda(\Delta\bar{x} * h) s^2$$

so that, using property (2.55), we get

$$\begin{aligned} \|A[D_x f(\bar{x}_s, \lambda_s) - D_x f(\bar{x}_0, \lambda_0)]\|_1 &= \sup_{\|h\|_1 \leq 1} \|A[D_x f(\bar{x}_s, \lambda_s) - D_x f(\bar{x}_0, \lambda_0)]h\|_1 \\ &\leq \sup_{\|h\|_1 \leq 1} \|A(\Delta\lambda h - 2\Delta\lambda(\bar{x}_0 * h) - 2\lambda_0(\Delta\bar{x} * h))\|_1 \\ &\quad + \sup_{\|h\|_1 \leq 1} \|A(2\Delta\lambda(\Delta\bar{x} * h))\|_1 \\ &\leq \|A\|_1 (|\Delta\lambda| + 2|\Delta\lambda|\|\bar{x}_0\|_1 + 2|\lambda_0|\|\Delta\bar{x}\|_1 + 2|\Delta\lambda|\|\Delta\bar{x}\|_1) \\ &=: \hat{Z}_0. \end{aligned}$$

To compute the bound Z_2 satisfying (3.7), let $c \in \overline{B_r(\bar{x}_s)}$. Then, applying the mean value inequality (see Corollary 2.2.6) and property (2.55) leads to

$$\begin{aligned} \|D_x f(c, \lambda_s) - D_x f(\bar{x}_s, \lambda_s)\|_1 &\leq \sup_{\xi \in \overline{B_r(\bar{x}_s)}} \|D_x^2 f(\xi, \lambda_s)\|_1 \|c - \bar{x}_s\|_1 \\ &= \sup_{\xi \in \overline{B_r(\bar{x}_s)}} \sup_{h_1, h_2 \in B_1(0)} \|-2\lambda_s(h_1 * h_2)\|_1 \|c - \bar{x}_s\|_1 \\ &\leq 2 \sup_{s \in [0,1]} |\lambda_s| r. \end{aligned}$$

This leads to the constant bound

$$Z_2 \stackrel{\text{def}}{=} 2\|A\|_1 \max_{s \in [0,1]} |\lambda_s|.$$

Using the above bounds, we define the radii polynomial as in (3.8) and applied the rigorous parameter continuation starting at $\lambda = 1.1$ and $\bar{x} \in \mathbb{R}^{N+1}$ with $\bar{x}_1 = 0.15$ and $\bar{x}_k = 0$ for all other components. We considered the different values $N = 2, 4, 6, 8, 10$, and this lead to five different branches denoted by $\mathcal{B}_2$, $\mathcal{B}_4$, $\mathcal{B}_6$, $\mathcal{B}_8$ and $\mathcal{B}_{10}$ (see Figure 3.6). We notice that as we increase N (the dimension of the problem), the branches $\mathcal{B}_N$ seem to converge. This is typical for Galerkin projections, which we will consider later in this book.

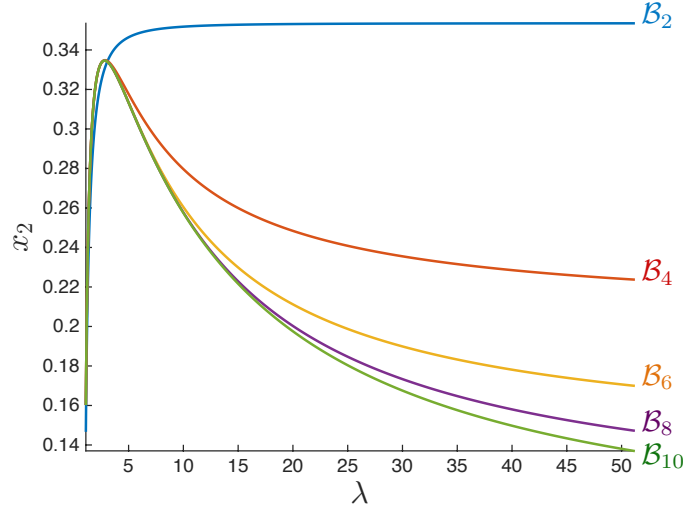


Figure 3.6: Computing global branches of solutions of $f(x, \lambda) = 0$ where $f : \mathbb{R}^{N+1} \times \mathbb{R} \rightarrow \mathbb{R}^{N+1}$ is given by (3.12) for different values of $N \in \{2, 4, 6, 8, 10\}$. Different projection dimensions N lead to different branches $\mathcal{B}_N$.

3.3 Rigorous Pseudo-Arclength Continuation

The method of Section 3.2 has the same disadvantage as the parameter continuation method (introduced in Section 3.1.1) in the sense that it will necessarily fail close to a fold, as the solution curve can not longer be parameterized by the natural parameter λ . We introduced in Section 3.1.2 the pseudo-arclength continuation as a way to overcome this problem. The goal of this section is to propose a rigorous pseudo-arclength continuation.

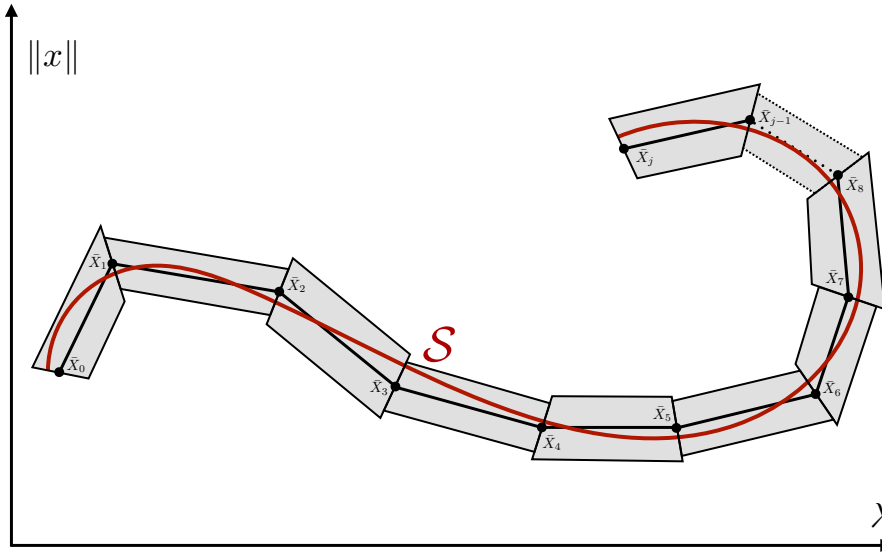


Figure 3.7: Computing a global solution branch via rigorous pseudo-arclength continuation.

As before, the process begins by assuming the existence of two numerical approximations $\bar{X}_0 = (\bar{x}_0, \lambda_0) \in \mathbb{R}^{n+1}$ and $\bar{X}_1 = (\bar{x}_1, \lambda_1) \in \mathbb{R}^{n+1}$ of the problem $f(X) = f(x, \lambda) = 0$, with $f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$. Let also assume that we have computed two approximate tangent vectors $\dot{X}_0$ and $\dot{X}_1$ of the solution curve at $\bar{X}_0$ and $\bar{X}_1$, respectively, that is $D_X f(\bar{X}_0) \dot{X}_0 \approx 0$ and $D_X f(\bar{X}_1) \dot{X}_1 \approx 0$.

$$\bar{X}_s = s\bar{X}_1 + (1-s)\bar{X}_0, \quad s \in [0, 1] \quad (3.13)$$

$$\dot{X}_s = s\dot{X}_1 + (1-s)\dot{X}_0, \quad s \in [0, 1]. \quad (3.14)$$

Finally, let $E_s(X) \stackrel{\text{def}}{=} (X - \bar{X}_s) \cdot \dot{X}_s$, and

$$F_s(X) \stackrel{\text{def}}{=} \begin{pmatrix} E_s(X) \\ f(X) \end{pmatrix}. \quad (3.15)$$

Denote by Π_s the hyperplane defined by the set of points X such that $E_s(X) = 0$, that is

$$\Pi_s \stackrel{\text{def}}{=} \left\{ X \in \mathbb{R}^{n+1} \mid (X - \bar{X}_s) \cdot \dot{X}_s = 0 \right\}.$$

The following theorem provides a mechanism to compute rigorously branches of solutions of $f(X) = 0$ for a smooth function $f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$. The idea is to prove the existence of $\tilde{X}(s) \in \mathbb{R}^{n+1}$ such that $F_s(\tilde{X}(s)) = 0$ for all $s \in [0, 1]$. Since the proof is similar to the one of Theorem 3.2.3, it is left as an exercise.

Theorem 3.3.1. *Consider $f \in C^k(\mathbb{R}^{n+1}, \mathbb{R}^n)$ with $k \in \{2, \dots, \infty, \omega\}$. For any $s \in [0, 1]$, consider the predictors $\bar{X}_s$ given by (3.13) and the tangent vectors $\dot{X}_s$ given by (3.14). Consider F_s given by (3.15). Let $A \in M_{n+1}(\mathbb{R})$ be such that $A \approx D_X F(\bar{X}_0)^{-1}$. Consider non negative bounds $Y_0, \hat{Y}_0, Z_0, \hat{Z}_0, Z_2$ satisfying*

$$\|AF_0(\bar{X}_0)\| \leq Y_0, \quad (3.16)$$

$$\|A[F_s(\bar{X}_s) - F_0(\bar{X}_0)]\| \leq \hat{Y}_0, \quad \text{for all } s \in [0, 1] \quad (3.17)$$

$$\|Id - AD_X F_0(\bar{X}_0)\| \leq Z_0, \quad (3.18)$$

$$\|A[D_X F_s(\bar{X}_s) - D_X F_0(\bar{X}_0)]\| \leq \hat{Z}_0, \quad \text{for all } s \in [0, 1] \quad (3.19)$$

$$\|A[D_X F_s(C) - D_X F_s(\bar{X}_s)]\| \leq Z_2(r)r, \quad \text{for all } C \in \overline{B_r(\bar{X}_s)} \text{ and } s \in [0, 1]. \quad (3.20)$$

Define the radii polynomial

$$p(r) \stackrel{\text{def}}{=} Z_2(r)r^2 - (1 - Z_0 - \hat{Z}_0)r + Y_0 + \hat{Y}_0.$$

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then there exists a function

$$\tilde{X} : [0, 1] \rightarrow \bigcup_{s \in [0, 1]} \overline{B_{r_0}(\bar{X}_s)}$$

with $\tilde{X} \in C^k((0, 1), \mathbb{R}^{n+1})$, and such that

$$f(\tilde{X}(s)) = 0, \quad \text{for all } s \in [0, 1].$$

Proof. The assumption that $p(r_0) = (Z_2(r_0)r_0 + Z_0 + \hat{Z}_0)r_0 - r_0 + Y_0 + \hat{Y}_0 < 0$ leads to

$$\kappa \stackrel{\text{def}}{=} Z_0 + \hat{Z}_0 + Z_2(r_0)r_0 < 1 - \frac{Y_0 + \hat{Y}_0}{r_0} \leq 1. \quad (3.21)$$

Define the Newton-like operator $T : \mathbb{R}^{n+1} \times [0, 1] \rightarrow \mathbb{R}^{n+1}$ by

$$T_s(X) = X - AF_s(X).$$

We begin by showing that for each $s \in [0, 1]$, the operator T_s is a contraction mapping from $\overline{B_{r_0}(\bar{X}_s)}$ into itself. Now, given $\xi \in \overline{B_{r_0}(\bar{X}_s)}$ and applying the bounds (3.18), (3.19) and (3.20), we obtain

$$\begin{aligned} \|D_X T_s(\xi)\| &= \|Id - AD_X F_s(\xi)\| \\ &\leq \|Id - AD_X F_0(\bar{X}_0)\| + \|A[D_X F_0(\bar{X}_0) - D_X F_s(\bar{X}_s)]\| \\ &\quad + \|A[D_X F_s(\xi) - D_X F_s(\bar{X}_s)]\| \\ &\leq Z_0 + \hat{Z}_0 + Z_2(r_0)r_0 = \kappa < 1, \end{aligned} \tag{3.22}$$

where we applied (3.21) to obtain the strict inequality. We now show that for each $s \in [0, 1]$ the operator T_s maps $\overline{B_{r_0}(\bar{X}_s)}$ into itself. Let $\xi \in \overline{B_{r_0}(\bar{X}_s)}$ and apply the Mean Value Inequality (Corollary 2.2.6) to obtain

$$\begin{aligned} \|T_s(\xi) - \bar{X}_s\| &\leq \|T_s(\xi) - T_s(\bar{X}_s)\| + \|T_s(\bar{X}_s) - \bar{X}_s\| \\ &\leq \sup_{C \in \overline{B_{r_0}(\bar{X}_s)}} \|D_X T_s(C)\| \|C - \bar{X}_s\| + \|AF_s(\bar{X}_s)\| \\ &\leq \sup_{C \in \overline{B_{r_0}(\bar{X}_s)}} \|D_X T_s(C)\| \|C - \bar{X}_s\| + \|AF_0(\bar{X}_0)\| + \|A[F_s(\bar{X}_s) - F_0(\bar{X}_0)]\| \\ &\leq (Z_0 + \hat{Z}_0 + Z_2(r_0)r_0)r_0 + Y_0 + \hat{Y}_0 \\ &< r_0, \end{aligned}$$

where the second to last inequality follows from (3.22) and the last inequality follows from $p(r_0) < 0$. Hence, for all $\xi \in \overline{B_{r_0}(\bar{X}_s)}$ and $s \in [0, 1]$, $T_s(\xi) \in \overline{B_{r_0}(\bar{X}_s)} \subset \overline{B_{r_0}(\bar{X}_s)}$.

Letting $\xi_1, \xi_2 \in \overline{B_{r_0}(\bar{X}_s)}$, apply (3.22) to obtain

$$\begin{aligned} \|T_s(\xi_1) - T_s(\xi_2)\| &\leq \sup_{C \in \overline{B_{r_0}(\bar{X}_s)}} \|D_X T_s(C)\| \|\xi_1 - \xi_2\| \\ &\leq (Z_0 + \hat{Z}_0 + Z_2(r_0)r_0) \|\xi_1 - \xi_2\| \\ &= \kappa \|\xi_1 - \xi_2\|. \end{aligned} \tag{3.23}$$

Define the operator

$$\begin{aligned} \mathcal{T} : \overline{B_{r_0}(0)} \times [0, 1] &\rightarrow \overline{B_{r_0}(0)} \\ (\xi, s) &\mapsto \mathcal{T}(\xi, s) \stackrel{\text{def}}{=} T_s(\xi + \bar{X}_s) - \bar{X}_s. \end{aligned}$$

Consider now $\xi_1, \xi_2 \in \overline{B_{r_0}(0)}$ and $s \in [0, 1]$. Then, since $\xi_1 + \bar{X}_s, \xi_2 + \bar{X}_s \in \overline{B_{r_0}(\bar{X}_s)}$, we use (3.23) to get

$$\|\mathcal{T}(\xi_1, s) - \mathcal{T}(\xi_2, s)\| = \|T_s(\xi_1 + \bar{X}_s) - T_s(\xi_2 + \bar{X}_s)\| \leq \kappa \|\xi_1 - \xi_2\|, \quad \text{for all } s \in [0, 1].$$

Since $\kappa < 1$, we conclude that $\mathcal{T} : \overline{B_{r_0}(0)} \times [0, 1] \rightarrow \overline{B_{r_0}(0)}$ is a uniform contraction. By the uniform contraction theorem (Theorem 3.2.2), there exists $G : [0, 1] \rightarrow \overline{B_{r_0}(0)}$ such that

$$\mathcal{T}(G(s), s) = G(s).$$

Since $f \in C^k(\mathbb{R}^{n+1}, \mathbb{R}^n)$, then $F_s \in C^k(\mathbb{R}^{n+1}, \mathbb{R}^{n+1})$ and then $\mathcal{T} \in C^k(\overline{B_{r_0}(0)} \times (0, 1), \overline{B_{r_0}(0)})$. By the uniform contraction theorem, $G \in C^k((0, 1), \overline{B_{r_0}(0)})$. Let

$$\tilde{X}(s) \stackrel{\text{def}}{=} G(s) + \bar{X}_s$$

so that for all $s \in [0, 1]$

$$T_s(\tilde{X}(s)) = T_s(G(s) + \bar{X}_s) = \mathcal{T}(G(s), s) + \bar{X}_s = G(s) + \bar{X}_s = \tilde{X}(s).$$

Note that $\tilde{X} \in C^k((0, 1), \bigcup_{s \in [0, 1]} \overline{B_{r_0}(\bar{X}_s)})$. Since $T_s(X) = X - AF_s(X)$, we get that

$$T_s(\tilde{X}(s)) = \tilde{X}(s) - AF_s(\tilde{X}(s)) = \tilde{X}(s).$$

By (3.22), $Z_0 = \|Id - AD_X F_0(\bar{X}_0)\| < 1$, which implies that A is an injective matrix. Hence,

$$F_s(\tilde{X}(s)) = (E_s(\tilde{X}(s)), f(\tilde{X}(s))) = 0, \quad \forall s \in [0, 1] \implies f(\tilde{X}(s)) = 0, \quad \forall s \in [0, 1]. \quad \square$$

Denote

$$\Delta \bar{X} \stackrel{\text{def}}{=} \bar{X}_1 - \bar{X}_0, \quad \Delta \dot{X} \stackrel{\text{def}}{=} \dot{X}_1 - \dot{X}_0.$$

Theorem 3.3.2. *Suppose that the hypotheses of Theorem 3.3.1 are satisfied with the norm $\|\cdot\|_\infty$, yielding a branch of solution curve $\mathcal{S}_0 \stackrel{\text{def}}{=} \bigcup_{s \in [0, 1]} \tilde{X}(s)$. If*

$$-\Delta \bar{X} \cdot \dot{X}_0 + r_0 \|\Delta \dot{X}\|_1 + \left| \Delta \bar{X} \cdot \Delta \dot{X} \right| < 0, \quad (3.24)$$

then $\mathcal{S}_0$ is a smooth curve, that is, for all $s \in [0, 1]$, $\frac{d\tilde{X}}{ds}(s) \neq 0$.

Proof. Considering the equality $E_s(\tilde{X}(s)) = (\tilde{X}(s) - \bar{X}_s) \cdot \dot{X}_s = 0$ and taking its derivative with respect to s and using (3.24) leads to

$$\begin{aligned} -\frac{d\tilde{X}}{ds}(s) \cdot \dot{X}_s &= -\Delta \bar{X} \cdot \dot{X}_s + (\tilde{X}(s) - \bar{X}_s) \cdot \Delta \dot{X} \\ &= -\Delta \bar{X} \cdot \dot{X}_0 + (\tilde{X}(s) - \bar{X}_s) \cdot \Delta \dot{X} - s \Delta \bar{X} \cdot \Delta \dot{X} \\ &\leq -\Delta \bar{X} \cdot \dot{X}_0 + \left| (\tilde{X}(s) - \bar{X}_s) \cdot \Delta \dot{X} \right| + \left| \Delta \bar{X} \cdot \Delta \dot{X} \right| \\ &\leq -\Delta \bar{X} \cdot \dot{X}_0 + \|\tilde{X}(s) - \bar{X}_s\|_\infty \|\Delta \dot{X}\|_1 + \left| \Delta \bar{X} \cdot \Delta \dot{X} \right| \\ &\leq -\Delta \bar{X} \cdot \dot{X}_0 + r_0 \|\Delta \dot{X}\|_1 + \left| \Delta \bar{X} \cdot \Delta \dot{X} \right| \\ &< 0, \end{aligned}$$

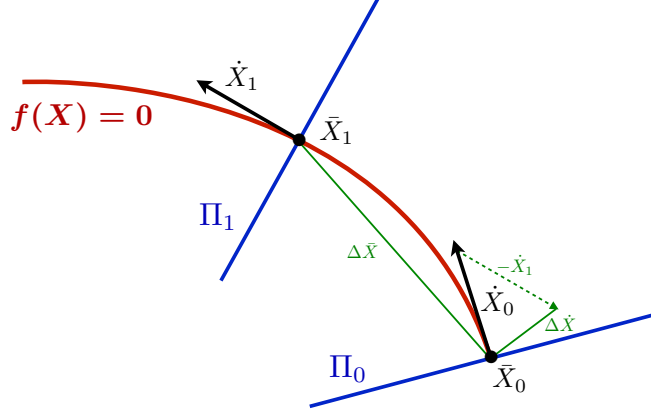


Figure 3.8: The important quantities involved in the hypothesis (3.24) of Theorem 3.3.2.

for all $s \in [0, 1]$. Hence, $\frac{d\tilde{X}}{ds}(s) \neq 0$ for all $s \in [0, 1]$, yielding the smoothness of $\mathcal{S}_0$. $\square$

Remark that the hypothesis (3.24) is reasonable if r_0 is small enough. See Figure 3.8 for a geometric representation of the important quantities involved in this hypothesis. One can see there that if the length $\Delta\tilde{X}$ of the segment $[\tilde{X}_0, \tilde{X}_1]$ is small, then the vectors $\Delta\tilde{X}$ and $\Delta\dot{X}$ should be close to be perpendicular and the vectors $\Delta\tilde{X}$ and $\dot{X}_0$ should be close to be parallel. Hence, $\Delta\tilde{X} \cdot \dot{X}_0$ should be close to the value $\|\Delta\tilde{X}\| \|\dot{X}_0\|$ while the vector $\Delta\tilde{X} \cdot \Delta\dot{X}$ should be close to 0. Hence, for small enough value of r_0 , $r_0 \|\Delta\dot{X}\|_1$ should be small enough so that inequality (3.24) is satisfied.

Assuming that two consecutive smooth curves have been rigorously computed, we now prove that they connect smoothly in one curve. For a proof, we refer to [5].

Theorem 3.3.3. *Assume that the hypotheses of Theorem 3.3.1 and of Theorem 3.3.2 are satisfied between $\tilde{X}_0$ and $\tilde{X}_1$ (yielding a smooth solution curve $\mathcal{S}_0$), and also between $\tilde{X}_1$ and $\tilde{X}_2$ (yielding a smooth solution curve $\mathcal{S}_1$). Then $\mathcal{S}_0 \cup \mathcal{S}_1$ is a smooth curve.*

Proof. We begin the proof by setting up some notation, differentiating the problems solved over the segments $[\tilde{X}_0, \tilde{X}_1]$ and $[\tilde{X}_1, \tilde{X}_2]$. Given $s \in [0, 1]$, denote

$$\begin{aligned} \bar{X}_s^{(0)} &= s\bar{X}_1 + (1-s)\bar{X}_0, & \dot{X}_s^{(0)} &= s\dot{X}_1 + (1-s)\dot{X}_0, \\ \bar{X}_s^{(1)} &= s\bar{X}_2 + (1-s)\bar{X}_1, & \dot{X}_s^{(1)} &= s\dot{X}_2 + (1-s)\dot{X}_1. \end{aligned}$$

Moreover, given $i \in \{0, 1\}$ and $s \in [0, 1]$ denote $E_s^{(i)}(X) \stackrel{\text{def}}{=} (X - \bar{X}_s^{(i)}) \cdot \dot{X}_s^{(i)}$,

$$\Pi_s^{(i)} \stackrel{\text{def}}{=} \left\{ X \in \mathbb{R}^{n+1} \mid (X - \bar{X}_s^{(i)}) \cdot \dot{X}_s^{(i)} = 0 \right\}$$

and

$$F_s^{(i)}(X) \stackrel{\text{def}}{=} \begin{pmatrix} E_s^{(i)}(X) \\ f(X) \end{pmatrix}.$$

Finally, denote by $p^{(0)}(r)$ (resp. $p^{(1)}(r)$) the radii polynomial built for the existence proof over the segment $[\bar{X}_0, \bar{X}_1]$ (resp. $[\bar{X}_1, \bar{X}_2]$). By hypothesis, since Theorem 3.3.1 is satisfied over the segment $[\bar{X}_0, \bar{X}_1]$ (resp. $[\bar{X}_1, \bar{X}_2]$), consider the radius $r^{(0)} > 0$ (resp. $r^{(1)} > 0$) satisfying $p^{(0)}(r^{(0)}) < 0$ (resp. $p^{(1)}(r^{(1)}) < 0$). Without loss of generality, one can assume that $r^{(0)} \neq r^{(1)}$ by continuity of the radii polynomials. Combining Theorem 3.3.1 and Theorem 3.3.2 gives existence of two smooth functions $\tilde{X}^{(0)} : [0, 1] \rightarrow \mathbb{R}^{n+1}$ and $\tilde{X}^{(1)} : [0, 1] \rightarrow \mathbb{R}^{n+1}$ whose images are $\mathcal{S}_0$ and $\mathcal{S}_1$. First we prove that the two curves connect, that is $\tilde{X}^{(0)}(1) = \tilde{X}^{(1)}(0)$. By construction, $\bar{X}_1^{(0)} = \bar{X}_1 = \bar{X}_0^{(1)}$, $\Pi_1^{(0)} = \Pi_0^{(1)}$, $E_1^{(0)} = E_0^{(1)}$ and therefore $F_1^{(0)} = F_0^{(1)}$. Moreover, $\tilde{X}^{(0)}(1)$ is the only solution of $F_1^{(0)}(X) = 0$ in $\overline{B_{r^{(0)}}(\bar{X}_1)}$ and $\tilde{X}^{(1)}(0)$ is the only solution of $F_0^{(1)}(X) = 0$ in $\overline{B_{r^{(1)}}(\bar{X}_1)}$. Since $F_1^{(0)} = F_0^{(1)}$, we conclude by unicity of the solutions that $\tilde{X}^{(0)}(1) = \tilde{X}^{(1)}(0)$.

Using a similar argument than in the proof of Lemma 3.2.4, there exists $\delta_0 > 0$ such that the proof over the segment $\{\bar{X}_s^{(0)} \mid s \in [0, 1]\}$ extends over the segment $\{\bar{X}_s^{(0)} \mid s \in [0, 1 + \delta_0]\}$. As a result, $\tilde{X}^{(0)}$ can be extended into a smooth function defined on $[0, 1 + \delta_0]$ such that $f(\tilde{X}^{(0)}(s)) = 0$, for all s in $[0, 1 + \delta_0]$. Hence, $\tilde{X}^{(0)}([1, 1 + \delta_0]) \subset \mathcal{S}_1$, and the proof follows. $\square$

The following result can be used to determine that a path of solution does not undergo any secondary bifurcations. For a proof, we refer to [5].

Corollary 3.3.4. *Assume that the hypotheses of Theorem 3.3.1 and Theorem 3.3.2 hold. Then for every $s \in [0, 1]$, $\dim \left(\text{Ker} \left(D_X f \left(\tilde{X}(s) \right) \right) \right) = 1$, that is $\tilde{X}(s)$ is a regular path.*

Proof. Given and $s \in [0, 1]$ and $\xi \in \overline{B_{r_0}(\bar{X}_s)}$, we get from (3.22) that $\|D_X T_s(\xi)\| < 1$, and so we have that

$$\|I - AD_X F_s(\tilde{X}(s))\| < 1,$$

which then implies that $D_X F_s(\tilde{X}(s))$ is invertible for all $s \in [0, 1]$. Let us show that for every $s \in [0, 1]$, $\dim \left(\text{Ker} \left(D_X f \left(\tilde{X}(s) \right) \right) \right) = 1$. Suppose by contradiction that there exists $s \in [0, 1]$ such that $\dim \left(\text{Ker} \left(D_X f \left(\tilde{X}(s) \right) \right) \right) > 1$. Let $U, V \in \mathbb{R}^{n+1}$ be two linearly independent non-trivial vectors such that $D_X f(\tilde{X}(s))U = D_X f(\tilde{X}(s))V = 0$. Since $D_X F_s(\tilde{X}(s))$ is invertible,

$$D_X F_s(\tilde{X}(s))U = \begin{pmatrix} D_X E_s(\tilde{X}(s)) \cdot U \\ D_X f(\tilde{X}(s))U \end{pmatrix} = \begin{pmatrix} \dot{X}_s \cdot U \\ 0 \end{pmatrix} \neq 0.$$

Hence, $\dot{X}_s \cdot U \neq 0$ and similarly $\dot{X}_s \cdot V \neq 0$. Then we can define $W = \frac{U}{\dot{X}_s \cdot U} - \frac{V}{\dot{X}_s \cdot V} \neq 0$, which is non zero because U and V are linearly independent vectors. We conclude that $D_X F_s(\tilde{X}(s)) W = 0$. This is a contradiction. Hence $\dim \left(\text{Ker} \left(D_X f(\tilde{X}(s)) \right) \right) \leq 1$. Differentiating $f(\tilde{X}(s)) = 0$ with respect to s leads to

$$D_X f(\tilde{X}(s)) \left(\frac{d\tilde{X}}{ds}(s) \right) = 0.$$

The proof of Theorem 3.3.2 implies that $\frac{d\tilde{X}}{ds}(s) \neq 0$ since (3.24) holds. Hence, for every $s \in [0, 1]$, $\dim \left(\text{Ker} \left(D_X f(\tilde{X}(s)) \right) \right) = 1$, meaning by definition that $\tilde{X}(s)$ is a regular path. $\square$

Example 3.3.5. Consider the problem of computing branches of steady states for the atmospheric circulation model introduced by Edward N. Lorenz in [24], that we refer to as the *Lorenz84* system,

$$\begin{cases} x'_1 = -\alpha x_1 - x_2^2 - x_3^2 + \alpha \lambda \\ x'_2 = -x_2 + x_1 x_2 - \beta x_1 x_3 + \gamma \\ x'_3 = -x_3 + \beta x_1 x_2 + x_1 x_3. \end{cases} \quad (3.25)$$

Let us fix $\alpha = 0.25$, $\beta = 4$ and $\gamma = 0.5$, and leave λ as a parameter. At these parameter values, equilibria of (3.25) are solutions of

$$f(x, \lambda) \stackrel{\text{def}}{=} \begin{pmatrix} -\frac{1}{4}x_1 - x_2^2 - x_3^2 + \frac{\lambda}{4} \\ -x_2 + x_1 x_2 - 4x_1 x_3 + \frac{1}{2} \\ -x_3 + 4x_1 x_2 + x_1 x_3 \end{pmatrix} = 0. \quad (3.26)$$

We now apply the rigorous pseudo-arclength continuation to this problem. Denote $X = (x, \lambda)$. Let two numerical approximations $\bar{X}_0 = (\bar{x}_0, \lambda_0) \in \mathbb{R}^4$ and $\bar{X}_1 = (\bar{x}_1, \lambda_1) \in \mathbb{R}^4$ of the problem $f(X) = f(x, \lambda) = 0$, with $f : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. Let also assume that we have computed two approximate tangent vectors $\dot{X}_0$ and $\dot{X}_1$ of the solution curve at $\bar{X}_0$ and $\bar{X}_1$, respectively, that is $D_X f(\bar{X}_0) \dot{X}_0 \approx 0$ and $D_X f(\bar{X}_1) \dot{X}_1 \approx 0$. We wish to use Theorem 3.3.1 to prove the existence of a segment of solutions in the solution set $\mathcal{S} = \{X = (x, \lambda) \in \mathbb{R}^4 \mid f(X) = 0\}$. For $s \in [0, 1]$, recall that $\bar{X}_s = (1-s)\bar{X}_0 + s\bar{X}_1$ and $\dot{X}_s = (1-s)\dot{X}_0 + s\dot{X}_1$. Denote $\Delta \bar{X} \stackrel{\text{def}}{=} \bar{X}_1 - \bar{X}_0 = (\Delta \bar{x}, \Delta \bar{\lambda})$ and $\Delta \dot{X} \stackrel{\text{def}}{=} \dot{X}_1 - \dot{X}_0 = (\Delta \dot{x}, \Delta \dot{\lambda})$ so that $\bar{X}_s = \bar{X}_0 + s\Delta \bar{X}$ and $\dot{X}_s = \dot{X}_0 + s\Delta \dot{X}$.

Let $A \approx D_X F_0(\bar{X}_0)^{-1}$. Compute the Y_0 bound satisfying (3.16). To compute $\hat{Y}_0$

satisfying (3.17), we use that the vector field is quadratic to get the expansion

$$\begin{aligned}
F_s(\bar{X}_s) - F_0(\bar{X}_0) &= \begin{pmatrix} E_s(\bar{X}_s) - E_0(\bar{X}_0) \\ f(\bar{X}_s) - f(\bar{X}_0) \end{pmatrix} \\
&= s \begin{pmatrix} 0 \\ D_X f(\bar{X}_0) \Delta \bar{X} \end{pmatrix} + \frac{s^2}{2} \begin{pmatrix} 0 \\ D_{XX} f(\bar{X}_0) (\Delta \bar{X}, \Delta \bar{X}) \end{pmatrix} \\
&= s \begin{pmatrix} 0 \\ D_x f(\bar{x}_0, \lambda_0) \Delta \bar{x} + \frac{\partial f}{\partial \lambda}(\bar{x}_0, \lambda_0) \Delta \bar{\lambda} \end{pmatrix} \\
&\quad + \frac{s^2}{2} \begin{pmatrix} 0 \\ -(\Delta \bar{x})_2^2 - (\Delta \bar{x})_3^2 \\ (\Delta \bar{x})_1(\Delta \bar{x})_2 - 4(\Delta \bar{x})_1(\Delta \bar{x})_3 \\ 4(\Delta \bar{x})_1(\Delta \bar{x})_2 + (\Delta \bar{x})_1(\Delta \bar{x})_3 \end{pmatrix}.
\end{aligned}$$

Hence, we set

$$\hat{Y}_0 = \left\| A \begin{pmatrix} 0 \\ D_x f(\bar{x}_0, \lambda_0) \Delta \bar{x} + \frac{\partial f}{\partial \lambda}(\bar{x}_0, \lambda_0) \Delta \bar{\lambda} \end{pmatrix} \right\|_\infty + \left\| A \begin{pmatrix} 0 \\ -(\Delta \bar{x})_2^2 - (\Delta \bar{x})_3^2 \\ (\Delta \bar{x})_1(\Delta \bar{x})_2 - 4(\Delta \bar{x})_1(\Delta \bar{x})_3 \\ 4(\Delta \bar{x})_1(\Delta \bar{x})_2 + (\Delta \bar{x})_1(\Delta \bar{x})_3 \end{pmatrix} \right\|_\infty.$$

Compute Z_0 satisfying (3.18). To compute $\hat{Z}_0$ satisfying (3.19), expand

$$\begin{aligned}
D_X F_s(\bar{X}_s) - D_X F_0(\bar{X}_0) &= \begin{pmatrix} D_X E_s(\bar{X}_s) - D_X E_0(\bar{X}_0) \\ D_X f(\bar{X}_s) - D_X f(\bar{X}_0) \end{pmatrix} \\
&= \begin{pmatrix} \dot{X}_s - \dot{X}_0 \\ D_X^2 f(\bar{X}_0)(s \Delta \bar{X}) \end{pmatrix} \\
&= s \begin{pmatrix} \Delta \dot{X} \\ D_X^2 f(\bar{X}_0) \Delta \bar{X} \end{pmatrix} \\
&= s \begin{pmatrix} (\Delta \dot{x})_1 & (\Delta \dot{x})_2 & (\Delta \dot{x})_3 & \Delta \dot{\lambda} \\ 0 & -2(\Delta \bar{x})_2 & -2(\Delta \bar{x})_3 & 0 \\ (\Delta \bar{x})_2 - 4(\Delta \bar{x})_3 & (\Delta \bar{x})_1 & -4(\Delta \bar{x})_1 & 0 \\ 4(\Delta \bar{x})_2 + (\Delta \bar{x})_3 & 4(\Delta \bar{x})_1 & (\Delta \bar{x})_1 & 0 \end{pmatrix},
\end{aligned}$$

and using that $|s| \leq 1$, set

$$\hat{Z}_0 = \left\| A \begin{pmatrix} (\Delta \dot{x})_1 & (\Delta \dot{x})_2 & (\Delta \dot{x})_3 & \Delta \dot{\lambda} \\ 0 & -2(\Delta \bar{x})_2 & -2(\Delta \bar{x})_3 & 0 \\ (\Delta \bar{x})_2 - 4(\Delta \bar{x})_3 & (\Delta \bar{x})_1 & -4(\Delta \bar{x})_1 & 0 \\ 4(\Delta \bar{x})_2 + (\Delta \bar{x})_3 & 4(\Delta \bar{x})_1 & (\Delta \bar{x})_1 & 0 \end{pmatrix} \right\|_\infty.$$

To compute Z_2 satisfying (3.20), consider $C \in \overline{B_r(\bar{x}_s)}$ and $C \stackrel{\text{def}}{=} B - \bar{X}_s \in \overline{B_r(0)}$, that is $\|B\|_\infty \leq r$. Then, using the expansion

$$D_X F_s(C) - D_X F_s(\bar{X}_s) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -2b_2 & -2b_3 & 0 \\ b_2 - 4b_3 & b_1 & -4b_1 & 0 \\ 4b_2 + b_3 & 4b_1 & b_1 & 0 \end{pmatrix},$$

we set

$$Z_2 \stackrel{\text{def}}{=} \left\| |A| \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 5 & 1 & 4 & 0 \\ 5 & 4 & 1 & 0 \end{pmatrix} \right\|_\infty.$$

Using the above bounds, define the radii polynomial by $p(r) = Z_2 r^2 - (1 - Z_0 - \hat{Z}_0)r + Y_0 + \hat{Y}_0$.

We applied the pseudo-arclength continuation method of Section 3.1.2 with constant step size $\Delta_s = 0.01$ at the initial point

$$\bar{X}_0 = \begin{pmatrix} \bar{x}_0 \\ \bar{\lambda}_0 \end{pmatrix} = \begin{pmatrix} -0.134445880127583 \\ -0.134445880127583 \\ -0.170597269494144 \\ 1/2 \end{pmatrix},$$

with tangent vector $\dot{X}_0$ satisfying $D_X f(X_0)\dot{X}_0 \approx 0$. At each step of the continuation, we build the radii polynomial and verified the existence of $r_0 > 0$ such that $p(r_0) < 0$. We performed 250 successful proofs of existence of solution branch segments. For each step, we verified (3.24). By Theorem 3.3.3, we obtain a global smooth branch of steady states of equation (3.26). The plot of the global branch can be seen on the left in Figure 3.9. In the right plot in Figure 3.9, we show the minimum radius r_- of the existence interval $\text{El}(p) = (r_-, r_+)$ for each of the proof.

3.4 Exercices

Exercise 3.4.1. Use pseudo-arclength continuation to compute a numerical approximation of the set

$$\{(x, y) : x^4 + y^4 = 1\}.$$

Exercise 3.4.2. Use parameter continuation to compute a branch of equilibria of the FitzHugh-Nagumo equation

$$x' = f(x) = \begin{pmatrix} x_1(x_1 - a)(1 - x_1) - x_2 \\ \epsilon(x_1 - \gamma x_2) \end{pmatrix},$$

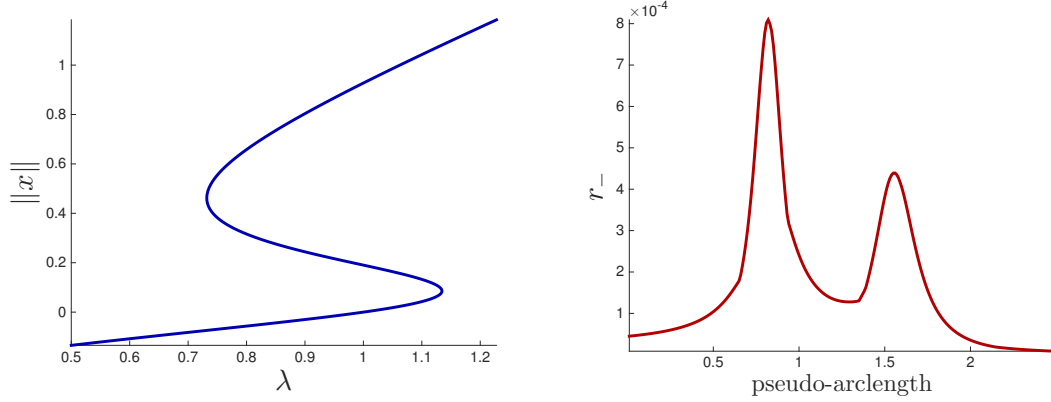


Figure 3.9: **(Left)** A branch of equilibria for the model (3.25) rigorously computed using the pseudo-arclength continuation algorithm as presented in Theorem 3.3.1. **(Right)** A plot of the smallest radius r_- from the existence interval for each proof as a function of the pseudo arclength.

with γ as the continuation parameter. Initiate the process at the parameter values $(a, \epsilon, \gamma) = (5, 1, 2)$. At each continuation step, decrease γ by 0.05. Repeat this procedure 40 times. What do you observe?

Exercise 3.4.3. Use pseudo-arclength continuation to compute a branch of equilibria of the FitzHugh-Nagumo equation

$$x' = f(x) = \begin{pmatrix} x_1(x_1 - a)(1 - x_1) - x_2 \\ \epsilon(x_1 - \gamma x_2) \end{pmatrix}.$$

Initiate the process at the parameter values $(a, \epsilon, \gamma) = (5, 1, 2)$, and leave γ variable so that $X = (x, \gamma)$. Begin the continuation at $X = (1.129171306613029, 0.564585653306514, 2)$ and make sure that the continuation goes in the direction of decreasing γ . Set $\Delta_s = 0.05$ and perform 500 steps of pseudo-arclength continuation.

Exercise 3.4.4. Use the rigorous parameter continuation of Section 3.2 to compute a branch of equilibria in the Lorenz equations (2.27). Fix the parameters $\sigma = 10$ and $\beta = 8/3$, and leave ρ as the continuation parameter. Initiate the process at $\rho_0 = 1.1$ at the point $\bar{x}_0 = (0.516397779494322, 0.516397779494322, 0.1)$. Using that there is a branch of equilibria of (2.27) parameterized over ρ is given by $x(\rho) = (\sqrt{\beta(\rho - 1)}, \sqrt{\beta(\rho - 1)}, \rho - 1)$, test the efficacy of the rigorous parameter continuation of Section 3.2 to identify the existence of branches of equilibria.

Exercise 3.4.5. Using the radii polynomials and the uniform contraction theorem, com-

pute a **branch** of period-3 orbits of the logistic map

$$F(x, \lambda) = \lambda x(1 - x)$$

starting at the orbit you computed rigorously in Exercise 2.7.15 at $\lambda = 3.83$. Prove the existence of the period-3 orbit for *all* parameter values $\lambda \in [3.8285, 4.25]$.

Exercise 3.4.6. Consider the problem $f : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}^2$, where

$$f(x, \lambda) = \begin{pmatrix} -x_1 + \lambda(1 - x_1)e^{x_2} \\ -3x_2 + 14\lambda(1 - x_1)e^{x_2} \end{pmatrix}, \quad \lambda \in \mathbb{R}. \quad (3.27)$$

At $\lambda = 0$, $\tilde{x} = (0, 0)$ satisfies $f(\tilde{x}) = 0$. Starting from this known solution, compute a branch of solution with the rigorous parameter continuation (as presented in Section 3.2). Can you get to $\lambda = 1$? Explain. Then, starting at the same point, apply the rigorous pseudo-arclength continuation (as presented in Section 3.3) to get to a solution at $\lambda = 1$.

Exercise 3.4.7. Define the map $f : \mathbb{R}^N \rightarrow \mathbb{R}^N$ given component-wise by

$$f_k(x) = (-k^4 + \lambda k^2 - \lambda \sigma) x_k - \lambda k^2 \sum_{\substack{k_1+k_2+k_3=k \\ k_i=-N, \dots, N}} x_{|k_1|} x_{|k_2|} x_{|k_3|}, \quad \text{for } k = 1, \dots, N,$$

where in the discrete convolution, it is assumed that $x_0 = 0$, and where $\lambda, \sigma \in \mathbb{R}$ are parameters. Fix the parameters $N = 10$, $\lambda = 10$ and $\sigma = 2$. Perform a rigorous parameter continuation in the parameter λ starting from $\bar{x} \in \mathbb{R}^{10}$ such that $\bar{x}_2 = 0.183635760099768$, $\bar{x}_6 = -0.002167639989372$ and $\bar{x}_{10} = 0.000023497621288$ with all other components equal to 0 to get a global smooth solution branch $\mathcal{B}_0 \in \mathbb{R}^{11}$. Go all the way to $\lambda = 250$, and plot the branch of solutions. Redo the same with $N = 20$ to get a global smooth solution branch $\mathcal{B}_1 \in \mathbb{R}^{21}$. Compare the solutions at the end of the branches $\mathcal{B}_0$ and $\mathcal{B}_1$. What do you observe? Plot $\mathcal{B}_0$ and $\mathcal{B}_1$ on top of one another.

Exercise 3.4.8. In Exercise 2.7.14, we studied the problem of existence of solutions of the problem $F(x, y) = 0$ where F is given by (2.57). In case of equal masses $m_1 = m_2 = m_3 = 1/3$, we showed that there are as many as ten solutions, and for the non equal masses case $m_1 = 0.9987451087$, $m_2 = 0.0010170039$ and $m_3 = 0.0002378873$, we showed that there were as many as eight solutions. Let $M_0 \stackrel{\text{def}}{=} (1/3, 1/3, 1/3)$, $M_1 \stackrel{\text{def}}{=} (0.9987451087, 0.0010170039, 0.0002378873)$, and

$$M_\lambda \stackrel{\text{def}}{=} \lambda M_1 + (1 - \lambda) M_0, \quad \text{for } \lambda \in [0, 1].$$

Consider now the parameter dependent problem

$$F(x, y, \lambda) \stackrel{\text{def}}{=} \begin{pmatrix} x - \sum_{i=1}^3 \frac{(M_\lambda)_i (x - x_i)}{\left(\sqrt{(x - x_i)^2 + (y - y_i)^2} \right)^3} \\ y - \sum_{i=1}^3 \frac{(M_\lambda)_i (y - y_i)}{\left(\sqrt{(x - x_i)^2 + (y - y_i)^2} \right)^3} \end{pmatrix} = 0.$$

For each of the ten points obtained rigorously in Exercise 2.7.14 at $\lambda = 0$ perform a rigorous parameter continuation in λ to try to get to $\lambda = 1$. Do you succeed for all points? Is that surprising? What might be happening?

Exercise 3.4.9. Prove Theorem 3.3.1.